FOREMOTION: LEARNING CAUSAL CO-SPEECH GESTURES THROUGH FUTURE-INFORMED SUPERVISION SUPPLEMENTARY MATERIAL

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This supplement provides additional details on ForeMotion's training objectives, reference implementation, dataset and evaluation protocol, and incremental generation interface. References to sections, equations, figures and tables in the main paper are marked explicitly. Code and sample output videos will be included in the accompanying supplementary archive.

S1 METHOD IMPLEMENTATION DETAILS

This section expands main-paper Section 3. Both predictive views use identical past/current speech windows and motion prefixes. The following definitions specify the token objectives; reference training schedules are given in Section S2.

S1.1 SPEECH-ACCESS BOUNDARIES AND ALIGNMENT

One gesture-token step corresponds to two motion frames, giving a 12.5-Hz token sequence for 25-fps motion. For a training clip starting at time u_{clip}, zero-based query t has timestamp $u_t = u_{\text{clip}} + (t+1)/12.5 - \epsilon_{\text{time}}$. The fixed offset $\epsilon_{\text{time}} = 10^{-9}$ seconds places the query just before its interval's right edge. The bounded policies locate a span satisfying $u_{\text{start}} \le u_t < u_{\text{end}}$. Word-bounded access uses word timestamps; segment-bounded access uses transcription segments. If malformed spans overlap, the narrowest enclosing interval is selected. When segment timestamps are unavailable, the utility can derive the segment extent from its valid word intervals.

For a selected span ending at u_{end}, the aligned cutoff is

$$B_\sigma(t) = \min(T-1, \max(t, \lceil 12.5(u_{\text{end}} - u_{\text{clip}})\rceil - 1)) \,. \tag{S1}$$

The span endpoint determines the lookahead boundary. If no span encloses the query, $B_\sigma(t) = t$, so the two views admit the same speech groups. Both retain the lower boundary $\max(0, t - L_s + 1)$ for every scope. The full-clip view uses $T-1$ as its upper boundary. All masks act on aligned audio/text groups; expanding an audio group into its code streams preserves the group's temporal boundary.

These bounds define the speech positions admitted by direct cross-attention. The two views share the lower boundary and gesture history; future conditioning extends the upper boundary to the enclosing word, enclosing transcription segment, or clip end.

S1.2 GESTURE TOKEN REGULARIZATION DETAILS

Gesture token regularization operates independently within each regional codec. The quantizer's input projection first maps the encoded latent into the initial residual. Let P_ℓ be the local level's comparison projection and $\mathbf{c}_{\ell,j}$ its stored code vector for entry $j \in \mathcal{V}$. At depth ℓ, distances are computed in the quantizer's comparison space:

$$d_{\ell,j} = \|P_\ell(\mathbf{r}_\ell) - \mathbf{c}_{\ell,j}\|_2^2, \qquad \Delta_{\ell,j} = d_{\ell,j} - d_{\ell,j_*}, \tag{S2}$$

where j_* is a nearest code. Let $\mathcal{N}_\ell$ contain the n nearest codes. For $n > 1$, the mean non-nearest distance gap defines the scale a_ℓ. With numerical floor $\varepsilon > 0$ and residual-target temperature $\tau_r > 0$,

the scale and target probabilities are

$$a_\ell = \max\left(\varepsilon, \frac{1}{n-1}\sum_{j\in\mathcal{N}_\ell\setminus\{j_*\}}\Delta_{\ell,j}\right), \qquad \pi_{\ell,j} = \frac{\exp[-\Delta_{\ell,j}/(a_\ell\tau_r)]}{\sum_{j'\in\mathcal{N}_\ell}\exp[-\Delta_{\ell,j'}/(a_\ell\tau_r)]}. \tag{S3}$$

Probability is zero outside $\mathcal{N}_\ell$; the implementation requires at least two candidates. The reference settings are $n = 8$, $\tau_r = 0.5$, and $\varepsilon = 10^{-8}$. At each valid position, sampling occurs with probability p_s and otherwise selects j_*. The chosen index is decoded back into residual space before applying main-paper Equation 4; the next depth's distances and targets are recomputed from that updated residual.

This adapts the distance-based soft labeling and stochastic quantization of RQ-Transformer (Lee et al., 2022) with regional restarts, a normalized nearest-code subset, partial sampling, and a hard/soft mixture. The codebook-distance targets depend on reference motion and the sampled residual path. The future-conditioned targets instead come from the neural generator under expanded speech access. These are distinct supervisory signals.

At step t, global stream k identifies a region and local depth: $\ell = k$ for $k < 8$, $\ell = k - 8$ for $8 \le k < 16$, and $\ell = k - 16$ otherwise. Thus $\widetilde{g}_{t,k}$, $\pi_{t,k,j}$ and $\mathcal{N}_{t,k}$ denote the corresponding regional selected code, target probability and candidate set; the batch index is omitted. With soft-target mixing weight $\beta \in [0, 1]$, the loss at one valid position in streams $k > 0$ is

$$\mathcal{L}^{\mathrm{RVQ}}_{t,k} = -(1-\beta)\log\bar{p}^{\mathrm{c}}_{t,k}(\widetilde{g}_{t,k}) - \beta\sum_{j\in\mathcal{N}_{t,k}}\pi_{t,k,j}\log\bar{p}^{\mathrm{c}}_{t,k}(j), \tag{S4}$$

where $\bar{p}^{\mathrm{c}}_{t,k}$ is the untempered training-head softmax over all $V + 1$ classes, including padding. The selected code in the hard term comes from the same residual-path construction as the soft distribution.

Canonical codec tokens supply temporal history and the hard target at global stream 0. Its sampled alternative can nevertheless enter the kinematic prefix. The sampled within-step vector supplies the kinematic prefix and hard targets at streams 1–19, including the first lower-body and face streams. It is cached once and reused for the detached pass. Unavailable regional targets remain padding, and input-only stream dropout remains an input mask. Validation token losses use deterministic codec targets.

S1.3 Loss reductions and shared training inputs

Each motion codebook contains $V = 2048$ entries. Each token head has $V + 1$ output classes, including padding. Both hard and soft token cross-entropy normalize over these classes, while padding targets are excluded. Let $\overline{\mathcal{L}}_k$ be the mean valid token loss for stream k, using canonical hard cross-entropy for $k = 0$ and Equation S4 otherwise. An all-invalid stream contributes zero. The token objective aggregates these losses as

$$\mathcal{L}_{\mathrm{token}} = \frac{1}{K}\sum_{k=0}^{K-1} w_k\overline{\mathcal{L}}_k, \qquad w_k = \begin{cases} 1, & 0 \le k < 16, \\ 2.5, & 16 \le k < 20. \end{cases} \tag{S5}$$

The denominator remains $K = 20$ when a regional stream is unavailable. By contrast, main-paper Equation 3 averages only over streams with valid targets and applies no facial multiplier. Its probability distributions exclude the padding logit.

The inherited regional latent contrastive objective compares reference quantized latents with latents reconstructed from straight-through Gumbel-softmax samples (Jang et al., 2017) of the causal logits, excluding the padding class. It pools nonoverlapping groups of 12 token steps, discards an incomplete final group, and normalizes the pooled vectors. For each region, reference vectors are queries and predicted vectors are candidates in a one-directional InfoNCE loss (van den Oord et al., 2018), with paired segments as positives, all pooled segments in the batch as candidates, and temperature 0.1. The three regional terms are summed.

The voice-activity head uses the temporal state and projected facial logits. Each pair of motion-frame labels is pooled to active when either frame is active. A sequence is eligible only if all its

original labels are valid. For BEATX, binary cross-entropy is summed over time and divided by the number of eligible sequences (with denominator offset 10^{-12}), rather than the number of valid time positions. The coefficients in main-paper Equation 5 apply to these reductions; their reference schedules are specified in Section S2.2.

Both predictive passes reuse the constructed speech inputs, including input-level modality dropout, and the same sampled kinematic prefixes. The future-conditioned prediction is detached when forming the distillation target, while gradients through the causal prediction update the shared generator.

S1.4 CAUSAL GENERATION

Generation executes the causal view, reuses attention caches, and feeds sampled gesture tokens into subsequent steps. The future-conditioned pass is used only during training. Section S4 describes the aligned speech inputs and incremental motion output.

S2 ARCHITECTURE AND REFERENCE IMPLEMENTATION

S2.1 REFERENCE CONFIGURATION AND INITIALIZATION

Table S1 summarizes the architecture and reference training configuration. The shared generator is initialized without resuming a gesture-generator checkpoint. Regional motion codecs are pretrained and frozen; audio/text embedding tables are initialized from Moshi and remain trainable in this configuration. Motion-token embedding tables are initialized from the regional codebooks and are also trainable. Both views use the same parameters, motion prefixes and past/current speech window, with only the future-conditioned view receiving later speech.

Table S1: Reference configuration settings, including the architecture, token representation, optimization parameters and coefficient schedules.

Component	Setting
Motion / audio rate	25 fps / 24 kHz
Training clip / token rate	10 seconds (250 frames) / 12.5 Hz (125 steps)
Regional motion streams	Upper body and hands: 8; lower body: 8; face: 4
Motion codebook size	2,048 entries per stream; one additional padding class in token heads
Temporal Transformer	4 layers, 2 attention heads; direct motion window: 25 steps
Kinematic Transformer	2 layers, 1 attention head
Direct causal speech window	$L_s = 50$ aligned speech groups
Distillation	$\tau_d = 1$; coefficient schedule in Equation S6
RVQ targets	$n = 8$, $\tau_r = 0.5$, $\varepsilon = 10^{-8}$
RVQ sampling / mixture	Maximum $p_s = 0.15$ / $\beta = 0.5$; Equation S7
Token stream weights	1 for body/hand streams; 2.5 for face streams
Auxiliary coefficients	Maximum $\lambda_v = 0.05$, $\lambda_c = 0.1$
Optimizer	AdamW; initial learning rate 10^{-4}; weight decay 0; gradient-norm clip 1
Precision / batch size	BF16 model, FP32 optimizer state / 128 sequences
Reference learning-rate schedule	Step decay by 0.8 every 2,000 epochs

In the reference implementation, Transformer dropout is 0.01. The optimizer uses AdamW (Loshchilov & Hutter, 2019) with its default momentum parameters.

S2.2 COEFFICIENT AND CONDITIONING SCHEDULES

Let e be the zero-based epoch index, i the zero-based minibatch index, and N_e the number of training minibatches in that epoch. Define fractional epoch $f = e + i/N_e$ and $\mathrm{clip}_{[0,1]}(x) = \min(1, \max(0, x))$. The reference distillation coefficient is

$$\lambda_d(e) = \begin{cases} 0, & e < 50, \\ 0.1 + 0.9\,\mathrm{clip}_{[0,1]}((e-50)/15), & e \geq 50. \end{cases} \tag{S6}$$

Thus the reference maximum is 1.0, reached at epoch 65. The no-distillation control uses $\lambda_d = 0$ throughout. Gesture token regularization uses

$$r(f) = \mathrm{clip}_{[0,1]}((f - 50)/50), \qquad p_s(f) = 0.15r(f), \qquad \beta(f) = 0.5r(f). \tag{S7}$$

When regularization is disabled, $p_s = \beta = 0$. These coefficients change sampling and token targets, not the number of RVQ levels or the gesture generator's architecture.

The reference auxiliary schedules use $\lambda_v = 0$ for $e \leq 50$ and 0.05 thereafter. The contrastive coefficient is zero for $e \leq 50$, $0.1i/N_e$ during epoch 51, and 0.1 thereafter. The straight-through Gumbel-softmax temperature is $1.0 - 0.6\,\mathrm{clip}_{[0,1]}((e - 50)/5)$ when the contrastive term is active. GAN and continuous reconstruction losses have zero weight.

During the first 50 epochs, speech conditioning is replaced by null tokens. Afterwards, the reference speech-memory dropout probability is $\max(0.1, 1 - (f - 50)/50)$. Independent lower-body and face input-stream dropout is enabled only for $e > 100$, with target fraction $\min(0.1, 0.1e/200)$. For each region, a fixed-size subset of eligible sequences is sampled independently, rounding its size down. These are input masks; missing regional targets remain excluded from their losses.

S2.3 Autoregressive generation

The generator begins from its dedicated initial token, then feeds its sampled gesture tokens back into the causal path. Cached attention states are reused across steps. The reference sampler uses nucleus sampling (Holtzman et al., 2020) at temperature 0.9 with temporal top-$p = 0.8$ and kinematic top-$p = 0.95$. Sampling settings, including classifier-free guidance, are configurable independently of the training-only future-conditioned pass. Section S4 describes the speech conditioning and incremental generation interface.

S3 Dataset and evaluation protocol

S3.1 Speaker selection and split membership

BEATX pairs speech and transcripts with mesh-based body, hand and facial motion (Liu et al., 2022; 2024). Following the 23-speaker protocol of Mughal et al. (2026), the multi-speaker setting excludes Carla and Itoi from the 25-speaker corpus. The single-speaker setting uses Scott. Evaluation includes 15 complete held-out utterances for Scott and 249 for the multi-speaker setting.

Partitioning is at the utterance level. The dataset builder reads explicit recording membership from the corpus split CSV, maps its additional-training entries to training and validation entries to development, and forms clips within the assigned partition. Training windows contain 250 motion frames (10 seconds); incomplete trailing windows are discarded. Evaluation uses complete utterances rather than the training windows. Motion is sampled at 25 fps, audio at 24 kHz, and the aligned gesture-token grid at 12.5 Hz.

S3.2 Baseline evaluation

We reevaluate released checkpoints for all baselines in main-paper Table 1 and train ForeMotion and its component/scope variants in this study. Table S2 records this distinction. The untreated first row of main-paper Table 2(a), with both components disabled, reuses the MIBURI evaluation from Table 1. It is a released-checkpoint reference, not a separately retrained MIBURI control. The remaining component and context-scope rows are variants trained in this study and evaluated in the single-speaker setting.

The single-speaker comparison includes all nine baselines; the multi-speaker comparison includes CaMN, EMAGE, MambaTalk, GestureLSM and MIBURI. Main-paper Table 1 groups the methods by their generation setting. DiffSHEG appears in the offline group, with its chunk-based streaming capability noted in the caption.

Table S2: Provenance of the reported comparisons. Baseline weights come from released checkpoints and are reevaluated; ForeMotion is trained in this study.

Method	Single	Multi	Weight source
HA2G	Yes	—	Released checkpoint
CaMN	Yes	Yes	Released checkpoints
TalkSHOW	Yes	—	Released checkpoint
DiffSHEG	Yes	—	Released checkpoint
EMAGE	Yes	Yes	Released checkpoints
MambaTalk	Yes	Yes	Released checkpoints
GestureLSM	Yes	Yes	Released checkpoints
PyraMotion	Yes	—	Released checkpoint
MIBURI	Yes	Yes	Released checkpoints
ForeMotion	Yes	Yes	Trained in this study

S3.3 Evaluation metrics

FGD compares generated and reference motion in a learned feature space, measuring the distance between the distributions through their feature means and covariances (Yoon et al., 2020). BeatAlign measures temporal correspondence between acoustic and kinematic beat events (Mughal et al., 2026). These metrics respectively assess distributional motion quality and synchronization with speech.

L1-Div measures within-sequence dispersion. For utterance u, let F_u be its number of evaluated frames and $\mathbf{y}_{u,r}$ its flattened joint-position vector at frame r. Then

$$\overline{\mathbf{y}}_u = \frac{1}{F_u}\sum_{r=1}^{F_u}\mathbf{y}_{u,r}, \qquad \text{L1-Div} = \frac{\sum_u \sum_{r=1}^{F_u} \|\mathbf{y}_{u,r} - \overline{\mathbf{y}}_u\|_1}{\sum_u F_u}. \tag{S8}$$

The L1 norm sums deviations over joint coordinates, and the denominator averages them over evaluated frames. Thus, L1-Div describes motion dispersion within each sequence; it does not measure variation across repeated generations for the same speech.

Facial-MSE measures mean-squared error between generated and reference facial motion. Lower FGD and Facial-MSE are preferred. For BeatAlign and L1-Div, we compare each score with its ground-truth counterpart, using $\rightarrow$ to indicate closeness to that value. Bold and underlined entries identify the best and second-best point estimates under these criteria. Together, the four metrics characterize complementary aspects of the generated motion.

S3.4 Perceptual protocol and response counts

The naturalness study uses 15 single-speaker test utterances rendered by six systems: ForeMotion, EMAGE, PyraMotion, MIBURI, CaMN and ground truth, yielding 90 videos. In each trial, the ForeMotion video is paired with another system's video for the same utterance; the comparator is chosen at random and system identity is concealed. Twenty-five participants answer the forced-choice question of which motion looks more natural.

Table S3: Single-speaker naturalness preferences. Selections count judgments favoring ForeMotion over the named comparator, divided by the number of judgments in that comparison. Rates are descriptive preferences pooled across the 25 participants and 15 utterances.

Comparator	EMAGE	PyraMotion	MIBURI	CaMN	Ground truth
Selections	38/42	25/33	27/38	26/40	14/39
Rate	90.5%	75.8%	71.1%	65.0%	35.9%

The counts sum to 192 judgments. Rates in Table S3 underlie main-paper Figure 3(a). They pool responses across utterances and participants and are reported descriptively. The study complements the automatic metrics with participants' assessments of motion naturalness for the same speech utterance.

S4 SPEECH CONDITIONING AND INCREMENTAL GENERATION

S4.1 AUDIO AND TRANSCRIPT PREPROCESSING

Evaluation uses cached audio codes and aligned transcript tokens. Mimi represents speech audio as discrete code indices on a 12.5-Hz grid. Transcript words are subword-tokenized and aligned to this grid using word-onset timestamps, with one text token per grid position and padding between text tokens. These timestamps specify alignment within recorded utterances, rather than the times at which a live speech recognizer releases the words.

The gesture generator initializes its lookup tables from Moshi's (Défossez et al., 2024) text and first eight audio embedding tables. At each aligned time step, the eight audio embeddings are summed, while the text token supplies a separate embedding. The *Moshi speech-text features* in main-paper Figure 2 therefore refer to these initialized token embeddings; no full-sequence contextual Moshi language model is run to produce these features.

S4.2 INCREMENTAL GESTURE GENERATION

At gesture step t, the generator receives the current aligned audio/text group and its retained past, produces a 20-stream gesture-token vector, and decodes two 25-fps motion frames. A dedicated start token initializes the first motion prediction; subsequent steps consume the generator's previously sampled tokens. It then updates its attention caches and advances by one token step. Its direct speech window contains at most 50 aligned groups, and its direct gesture-history window contains at most 25 steps. Both training views use identical past/current windows; the training-only view adds speech positions through $B_\sigma(t)$ as defined in main-paper Equation 2. Only the causal view is called during generation, so the training-only extension does not alter the generator's inference-time speech window.

The experiments evaluate this autoregressive generator at the supplied aligned-token interface, where the causal mask restricts direct speech attention to permitted past/current positions. End-to-end causality additionally requires prefix-stable upstream features, so that later audio or text cannot revise already supplied tokens, and a token-release and buffering policy that respects their availability. Each token step represents 80 ms of motion time; this interval is distinct from wall-clock processing latency and does not include speech recognition or input buffering.

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